

SMART CANTEEN MANAGEMENT & DEMAND ANALYSIS SYSTEM

Java Programming Project Report

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Project Type	Java / Object-Oriented Programming

A console-based Java application for canteen ordering, billing, menu management and basic demand analysis.

Project Report

Abstract

The Smart Canteen Management & Demand Analysis System is a Java-based console application designed to organize common college canteen activities. The system provides separate student and administrator operations, including login, menu viewing, food selection, order creation, billing, menu management and basic demand analysis. The implementation uses multiple Java classes to demonstrate object-oriented programming, modular design, collections, input validation and exception handling.

The project focuses on providing a simple and understandable implementation that can be compiled and executed from the terminal. The design diagrams in this report describe the users, workflow, sequence of operations and main classes used by the application.

1. Introduction

College canteen operations involve repeated activities such as displaying available food, accepting orders, calculating prices and keeping track of demand. When these activities are handled manually, repetitive calculations and input mistakes can occur. The proposed system provides a simple computerized approach using Java.

The application is intentionally console-based so that the project remains focused on Java programming and object-oriented concepts while still representing a realistic canteen workflow.

2. Problem Statement

A college canteen needs a simple way to manage food items and student orders while reducing manual work in menu handling and billing. The system should allow students to view food items, select quantities and receive a calculated bill. It should also provide administrators with menu-management functions and basic demand-related information.

3. Objectives

- Implement a student and admin login flow.
- Display available food items with their prices and availability.
- Allow students to select food items and quantities.
- Create orders using separate order and order-item classes.
- Calculate and display the total bill.
- Provide menu-management functions for the administrator.
- Provide basic demand-analysis output.
- Demonstrate object-oriented programming and modular Java design.
- Handle invalid user input safely.

4. Requirements Analysis

4.1 Functional Requirements

ID	Requirement	Description
FR1	Student Login	Authenticate student access.
FR2	Admin Login	Authenticate administrator access.

ID	Requirement	Description
FR3	Menu Display	Display food items, prices and availability.
FR4	Food Selection	Accept food ID and quantity.
FR5	Order Creation	Create and manage selected order items.
FR6	Billing	Calculate and display final order amount.
FR7	Demand Analysis	Provide basic demand-related information.
FR8	Validation	Handle invalid choices and quantities.

4.2 Non-Functional Requirements

- Usability: The console interaction should be simple and easy to understand.
- Reliability: Order and bill calculations should be consistent.
- Maintainability: Responsibilities should be divided among classes.
- Error Handling: Invalid input should be handled without unexpected termination.
- Extensibility: The modular design should support future GUI or database additions.

5. System Design

The system uses a modular object-oriented design. Main.java controls the program flow and connects the major modules. User contains common user behavior, while Student and Admin represent role-specific operations. FoodItem and Menu manage food information. Order and OrderItem manage order data. Billing calculates the bill and DemandAnalysis processes order-related information for basic analysis.

Module	Main Responsibility
Main	Controls application flow and console interaction.
User / Student / Admin	Provides common and role-specific user operations.
FoodItem / Menu	Stores and manages food information.
Order / OrderItem	Stores selected items and order details.
Billing	Calculates and displays order total.
DemandAnalysis	Provides basic demand-related analysis.

5.1 Use Case Diagram

USE CASE DIAGRAM — SMART CANTEEN

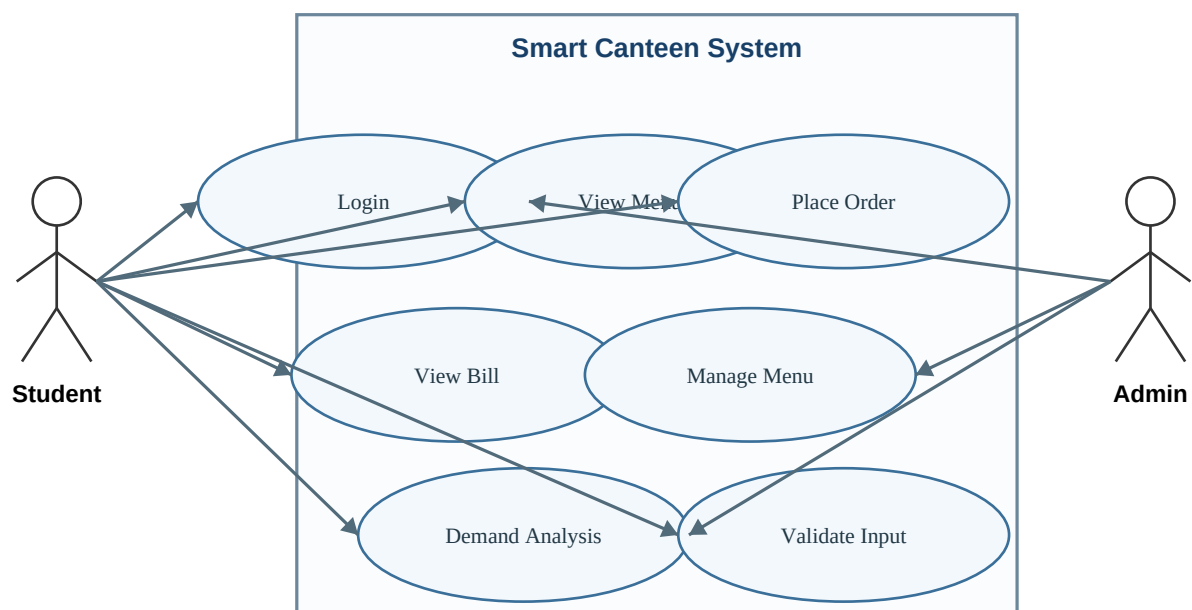


Figure 1. Main interactions of Student and Admin with the Smart Canteen System.

5.2 Workflow / Process Flow

SYSTEM WORKFLOW / PROCESS FLOW

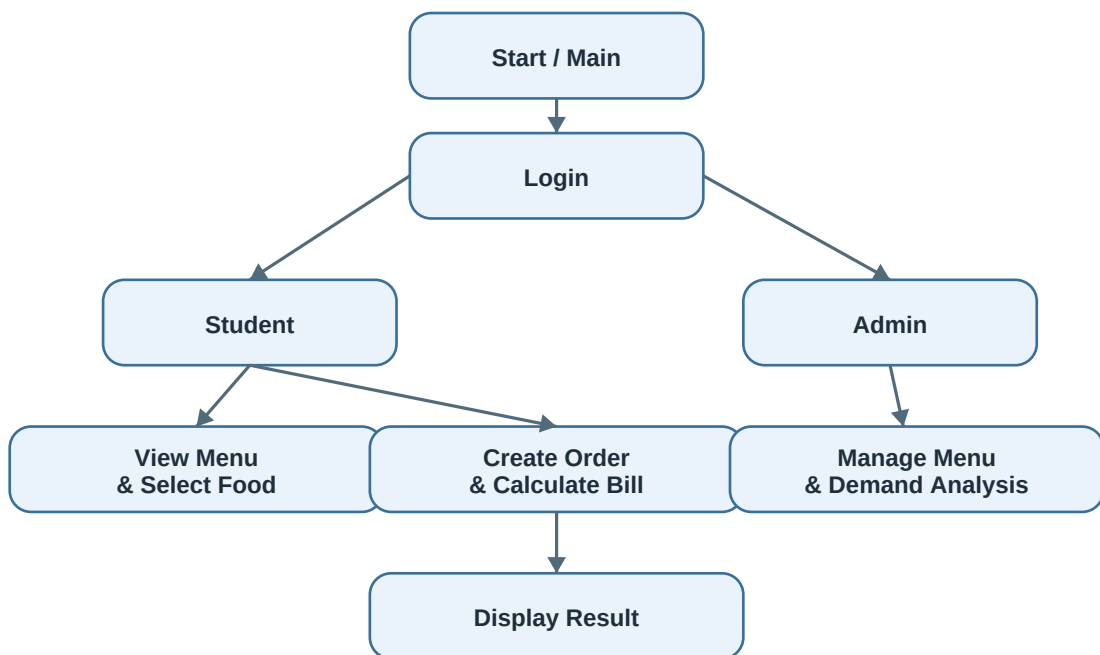


Figure 2. High-level workflow from login to student ordering/billing or admin operations.

5.3 Sequence Diagram

SEQUENCE DIAGRAM — ORDER PROCESS

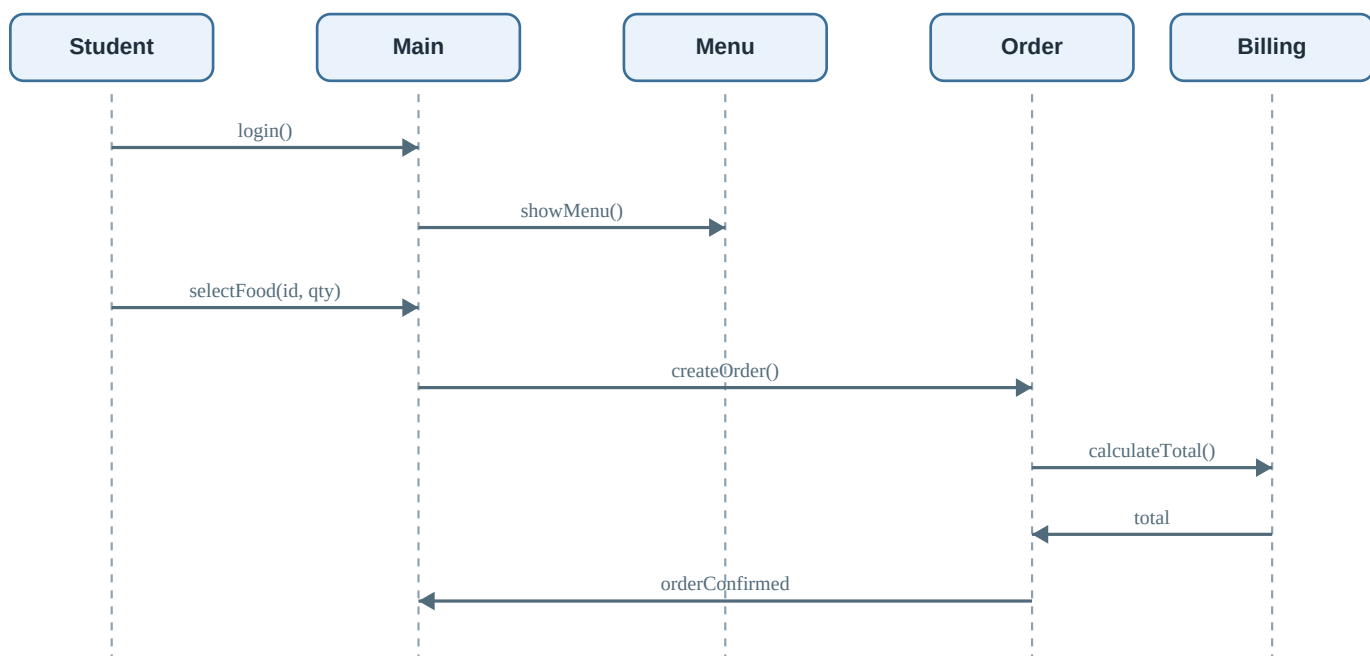


Figure 3. Sequence of interactions during a typical student order.

5.4 Class / Component Diagram

CLASS / COMPONENT DIAGRAM

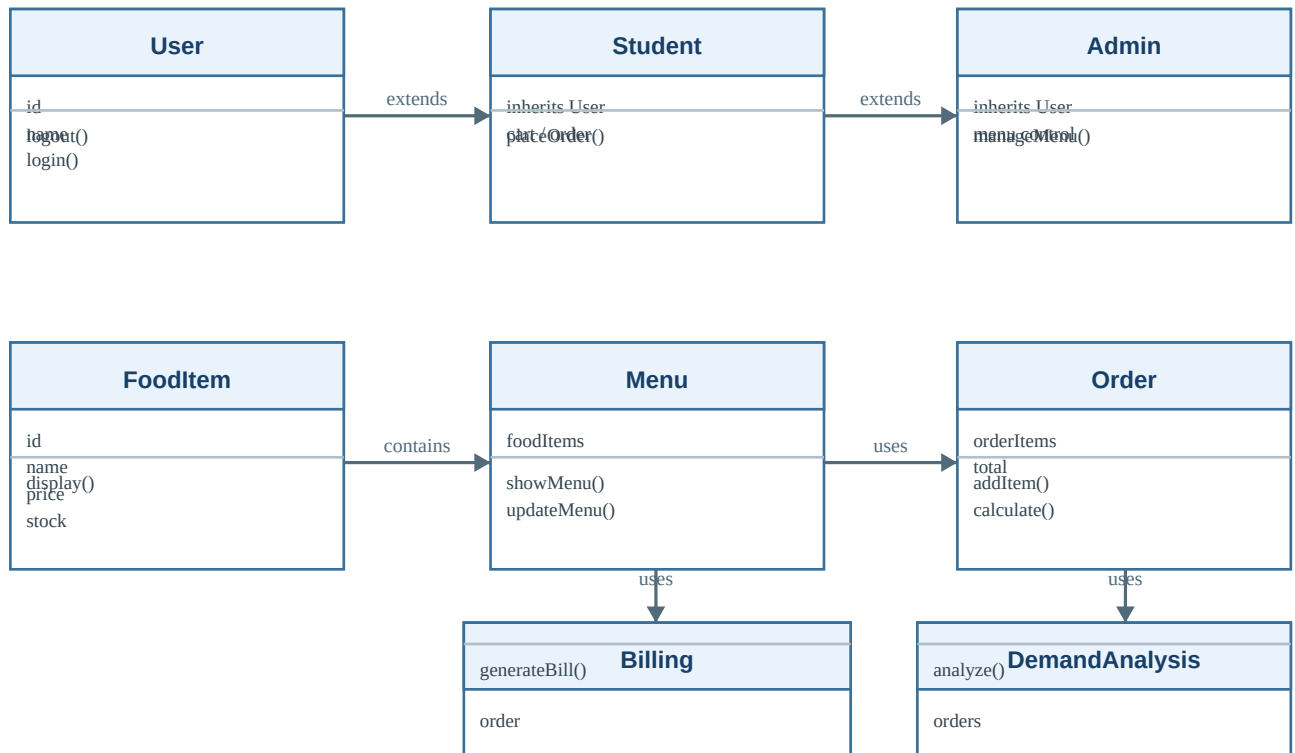


Figure 4. Main Java classes and their relationships.

6. Design Decisions & Rationale

Decision	Rationale
Modular class design	Each class has a focused responsibility, making the project easier to understand and maintain.
Inheritance	Student and Admin can reuse common User behavior.
ArrayList	Dynamic collections are suitable for food items and order items.
Input validation	Prevents or handles invalid food IDs, quantities and menu choices.
Exception handling	Helps the application handle invalid input without unexpected termination.
Console interface	Keeps the implementation focused on Java and OOP concepts.

7. Implementation Details

The application is organized into Java source files under the canteen package. Main.java is used for program flow and user interaction. The remaining classes separate user management, food/menu management, ordering, billing and demand analysis.

- Main.java — controls the overall application flow.
- User.java — stores common user information and login-related behavior.
- Student.java — implements student-specific functions such as placing orders.

- Admin.java — implements administrator functions such as menu management and analysis.
- FoodItem.java — represents an individual food item.
- Menu.java — manages the available food items.
- Order.java — represents an order and its selected items.
- OrderItem.java — stores an individual item and quantity within an order.
- Billing.java — calculates and displays the order amount.
- DemandAnalysis.java — provides basic demand/sales-related analysis.

7.1 Technologies Used

- Java programming language
- Object-Oriented Programming concepts
- ArrayList / Java Collections
- Exception handling and input validation
- Visual Studio Code / terminal-based execution

8. Testing & Results

Testing was performed by executing the main user flows and checking the console output. The tests cover login, menu access, food selection, quantity entry, order creation, billing and administrator analysis.

Test Case	Expected Result	Status
Valid student login	Student functions become available.	Tested
Valid admin login	Admin functions become available.	Tested
View menu	Available food items are displayed.	Tested
Select food and quantity	Valid selection is accepted.	Tested
Invalid input	Input is rejected or handled safely.	Tested
Create order	Selected items are stored in the order.	Tested
Bill calculation	Final amount is displayed.	Tested
Demand analysis	Basic demand-related output is displayed.	Tested

8.1 Result

The completed console application supports the intended canteen workflow: users can authenticate, access the appropriate menu, select food and quantity, create an order, and obtain billing output. Administrator functionality includes menu-related operations and demand-analysis output.

9. Challenges Faced

- Maintaining consistent communication between multiple Java classes.
- Organizing the project into a clear package and source-file structure.
- Handling incorrect choices and quantities entered through the console.
- Connecting order creation with billing and demand-analysis modules.
- Testing the application through repeated executions and different input cases.

10. Learnings & Key Takeaways

- Practical understanding of classes, objects and constructors.
- Use of inheritance for reusing common functionality.
- Use of ArrayList for dynamic data storage.
- Importance of modular design and separation of responsibilities.
- Importance of input validation and exception handling.
- Experience compiling and executing a multi-class Java project.

11. Future Enhancements

- Develop a graphical user interface for students and administrators.
- Add database storage for users, food items and orders.
- Add digital payment support.

- Generate richer sales and demand reports.
- Add stronger authentication and role-based access control.
- Add automated unit and integration tests.

12. Conclusion

The Smart Canteen Management & Demand Analysis System demonstrates how Java and object-oriented programming can be used to model a practical college canteen workflow. The modular class structure separates responsibilities for users, menu items, orders, billing and analysis. The resulting console application provides a clear foundation that can later be extended with persistent storage, a graphical interface and additional reporting features.

13. References

- VITyarthi — Build Your Own Project instructions and submission guidelines supplied for the project.
- Java programming language and standard library documentation.
- Project source code developed for the Smart Canteen Management & Demand Analysis System.